Multimedia Tools and Applications (2024) 83:90069–90088
https://doi.org/10.1007/s11042-024-19070-6

A novel medical image enhancement algorithm based on CLAHE and pelican optimization

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Received: 15 June 2023 / Revised: 22 December 2023 / Accepted: 22 March 2024 /
Published online: 5 April 2024

Abstract

Medical image enhancement is considered a challenging image-processing framework because the low quality of images resulting after acquisition and transmission seriously affects the clinical diagnosis and observation. In order to improve the medical image visual quality, a novel medical image enhancement algorithm that is based on contrast adaptive histogram equalization and pelican optimization algorithm is proposed in this work. The estimation process using our proposed model improves the efficiency of the operation and provides superior results in terms of image quality and contrast. There are three steps in the enhancement process. The primary step includes medical image generation using a Text-to-image generative model. Secondly, the estimation of the clip-limit, which controls the enhancing performance. Finally, the operation of enhancing the medical images using our proposed method. The simulation experiments prove that our proposed algorithm achieves superior performance qualitatively and quantitatively, compared with the state-of-the-art experimental methods, Upon a thorough examination and comparative analysis of performance parameters. Furthermore, the advantageous characteristic of this algorithm is its applicability in multiple types of images. Improving the quality of the medical images using our algorithm allows us to attain a superior visual impact on the processed image, and to increase the rate of conformity in the clinical diagnosis. Our proposed model illustrates the structure and forms of relevant details, contained in the medical images. This leads to an increase in overall contrast and enhances visual perception.

Keywords Bio-inspired optimization algorithm · Text-to-image diffusion · Image generation · Image enhancement · Contrast estimation

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1 Introduction

Image enhancement is the main pre-processing stage in different computer vision applications. The objective of this step is to improve the quality of the image, and subsequently increase the interpretability and the perception of the information [1]. In medical image enhancement techniques, the operational process is more challenging, due to acquisition artifacts, image fuzziness, and the heterogeneity of the brightness and noise levels [2, 3].

The main principle of an enhancement technique is to transform the charactical parameters of the image into a superior form. Contrast is deemed the most substantial parameter in the image quality evaluation procedure. It is generated by the luminance reflection of two adjacent regions, and it represents the difference in the visual proprieties, which allows an object to be distinguishable from other forms and surroundings. By utilizing image enhancement techniques, the low contrast issue will be solved by reinforcing the region of interest [4, 5].

Several enhancement techniques have been previously conducted to achieve more effective visualization of the image. These methods are classified based on their processing impact as spatial and transform domain techniques. Histogram equalization (HE) is one of the conventional spatial domain methods. This technique increases the image gray distribution's uniformity. Adaptive histogram equalization (AHE) is a classic HE-based method that is applied for local regions and it is effective in terms of reducing the loss of image details. However, the shortcoming of this technique is noise over-amplification [6].

The contrast-limited adaptive histogram equalization (CLAHE) diverges in the histogram distribution's operation, which is not the case with classical adaptive histogram equalization. CLAHE is an efficient algorithm to reinforce the local details of an image. The main limitation of this method is that the contrast reinforcement is restricted by clipping the histogram at a predefined clip (contrast) limit. On this basis, we present a novel method for medical image enhancement named CLAHE-POA, based on applying both CLAHE and pelican optimization algorithm (POA). The model proposed is salutary for estimating the most accurate contrast-limit value to achieve a superior enhancement performance.

In the recent period, metaheuristic algorithms applications are well known for solving different types of engineering optimization problems. These algorithms are essentially inspired by biological, physical, and chemical systems in nature, and have the ability to provide and find solutions in several problem areas. One of the nature-inspired meta-heuristic algorithms is the pelican optimization algorithm (POA). This model was able to solve the majority of engineering issues, and it was first introduced in solving image-processing problems [7–9].

The overall objective of this paper is to present a novel medical image enhancement method. This model is based on using POA to estimate the clip-limit, which controls the performance of the enhancement operation using CLAHE. The estimation process improves the efficiency of the operation and provides superior results in terms of image quality and contrast. The utilization of the present algorithm allows gaining a superior visual impact on the processed image as well as increasing the conformity rate in the clinical diagnosis.

The rest of the paper is organized as follows: Section 2 introduces the text-to-image stable diffusion model utilized for medical image generation, in addition to giving details about CLAHE and POA methods. The proposed approach and algorithm are given in Section 3. The experimental results are presented in Section 4. In Section 5, the analysis of the work and discussion are given, Conclusion of the proposed work is given in Section 6.

2 Related work

Many image enhancement frameworks have been developed based on the application of the CLAHE model, Diksha et al. [10] proposed an approach for underwater image enhancement using CLAHE combined with percentile methodologies, the performance of their proposed system surpasses the classical techniques in reinforcing this form of images. Ruiquin et al. [11] have presented an intelligent enhancement scheme based on CLAHE and F-Shirt transformation during decompression, their model balanced the enhancement effect while increasing the contrast and preserving the image details.

Kuran et al. [12] have proposed determining optimal parameters for CLAHE to enhance the contrast of images, based on using a Multi-objective Cuckoo Search. This model can quickly find the optimal solution in a relatively short time. However, it has limitations, including noise amplification caused by the inappropriate selection of the clip limit due to speeding up the iterations' number.

Novel applications of CLAHE focus essentially on utilizing it in pre-processing for classification and prediction tasks. Alwakid et al. [13] have presented a conjunction model based on CLAHE and enhanced super-resolution generative adversarial network (ESRGAN) for the diagnosis of all five stages of diabetic retinopathy (DR). Their proposed methodology outperformed the established models for detection [14].

Regarding utilizing the CLAHE model in the medical and biomedical domain, the enhancement of retinal fundus images using this method is a renowned application; Sarika et al. [15] have proposed a modified version of CLAHE. Their model resolves the noise amplification issue. However, the operational required time is remarkable [16–19].

Concerning POA, it is essentially utilized in engineering applications. Trojovský et al. [20] presented the fundamental principle of POA and its modelling. In addition to the comparison of its performance with additional competitor algorithms. A hybrid approach of POA is conducted in several works in the research literature. Rajam et al. [21] studied the improvement of the performance of grid-connected PV using the GBDT-POA approach. their proposed model provides optimal outcomes, with increasing the quality of the power. Furthermore, Kumar et al. [22] proposed a novel control technique using hybrid POA, and the optimum execution of accuracy resulted is superior compared to the existing techniques.

3 Materials and methods

In this section, we outline the steps and methods forming our proposed approach.

3.1 Medical image generation using text-to-image stable diffusion

Image generative applications focus on solving the inverse problem represented as the process of computing an unknown physical quantity from the measurement obtained via a direct model. Forward and inverse issues can be presented with the following expression,

$$\begin{aligned} A : X = Y \\ y = A(x) + e \end{aligned} \tag{1}$$

Where x ∈ X is the image, y ∈ Y is the data corrupted by the noise e [23–25].

The text-to-image generative model is defined as taking the natural language text description as input, along with generating an image to match that description. In the recent period, text-to-image models have significantly developed in terms of authenticity and originality. Stability AI's stable diffusion is an image generation model that offers XLA compilation and mixed precision. Various implementations that provide image-generation models from textual prompts are existed. However, the application of this method provides powerful advantages, including the attainment of state-of-the-art generation speed (Fig. 1).

The stable diffusion model is composed of three parts:

- Text-encoding,
- Image denoising using the diffusion model.
- Image decoding to achieve higher resolution [26–29].

The preliminary stage of our process involves text-to-image generation. Implementing the text-to-image generative model facilitates the acquisition of the targeted medical data essential for the examination and analysis of our enhancement algorithm. In addition,

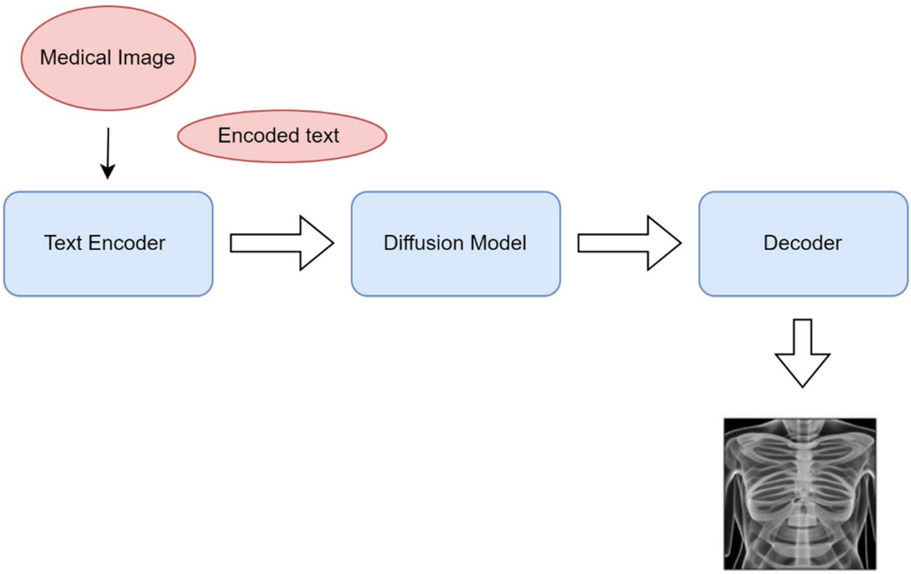


Fig. 1 The architecture of the text-to-image stable diffusion generative model

Our study includes a comprehensive evaluation of the versatility of our model, specifically examining its applicability to diverse datasets, including real medical images.

3.2 Contrast limited adaptive histogram equalization

CLAHE is a renowned method used in enhancing the low contrast problem in digital images. The performance of CLAHE in medical imaging applications outweighs adaptive histogram equalization (AHE) and classical histogram equalization (HE). CLAHE has two main features: the limitation of histograms' distribution to restrain the undue enhancement of noise spots on the one hand, and the acceleration of equalization using interpolation on the other hand.

The clip limit controls the performance of the CLAHE method, while the histograms of each non-overlapping region converge to a level below this parameter. The clip limit β form can be expressed as:

$$\beta = \frac{MN}{G}\{1 + \frac{\alpha}{100}(AS_{max} - 1)\} \tag{2}$$

Where M and N are the numbers of pixels in each region, G is the number of grayscale levels, α is the clip factor, and AS_{max} represents the maximum allowance slope. Based on the aforementioned details, the estimation of β value is crucial to achieve an optimal image quality (Fig. 2).

The steps of the CLAHE process are as follows:

- Splitting the image into different regions (Continuous/non-overlapping).
- Clipping the histogram of each region using the threshold.
- Reallocating pixel values, and distributing them evenly.
- Performing a local equalization in the regions.
- Reconstructing pixel values using linear interpolation.

3.3 Pelican optimization algorithm

POA is a new meta-heuristic optimization algorithm inspired by pelican behavior during foraging. In population-based algorithms, each member signifies a candidate solution. The population members' initialization is defined with the following equation:

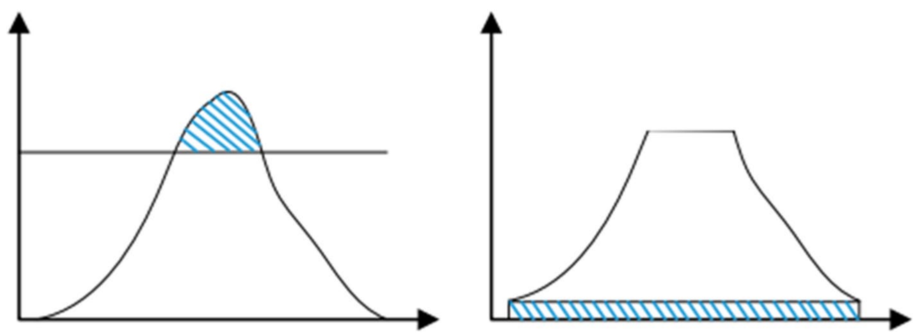


Fig. 2 The histogram's distribution before and after clipping

$$x_{i,j} = l_j + rand.\left(u_j - l_j\right), i = 1, 2, \ldots, N, j = 1, 2, \ldots, m, \tag{3}$$

Where $x_{i,j}$ is the value of variable jth and it is identified by the ith candidate solution, N represents the number of variables, m is the number of problems, *rand* is the random number in the interval [0,1], lj and uj are the lower and the upper bounds of the problem variables [30].

The population matrix identifies the members of pelicans in POA. It is given as:

$$X = \begin{bmatrix} X_1 \\ \cdot \\ X_i \\ \cdot \\ X_N \end{bmatrix}_{N*m} = \begin{bmatrix} x_{1,1} & x_{1,j} & x_{1,m} \\ x_{i,1} & x_{i,j} & x_{i,m} \\ x_{N,1} & x_{N,j} & x_{N,m} \end{bmatrix}_{N*m} \tag{4}$$

X is the population matrix and X_i is the ith of the pelican. Regarding the objective function values, they are obtained using the following equation:

$$F = \begin{bmatrix} F_1 \\ \cdot \\ F_i \\ \cdot \\ F_N \end{bmatrix}_{N*1} = \begin{bmatrix} F\left(X_1\right) \\ \cdot \\ F\left(X_i\right) \\ \cdot \\ F\left(X_N\right) \end{bmatrix}_{N*1} \tag{5}$$

F is the objective function vector. The hunting strategy of POA is composed of an exploration phase where the proposed POA discovers several areas of the researching space; and the exploitation phase that leads to the convergence of POA to a better solution in the hunting area. The best candidate solution acquired after the algorithm iterations is the optimal one for the given problem [31, 32].

The pelican strategy in moving towards the prey's environment is defined with the following equation:

$$x_{i,j}^{P_1} = \begin{cases} x_{i,j} + \text{rand} \cdot \left(p_j - I \cdot x_{i,j}\right), & F_p < F_i; \\ x_{i,j} + \text{rand} \cdot \left(x_{i,j} - p_j\right), & \text{else} \end{cases} \tag{6}$$

$x_{i,j}^{P_1}$ is the new status of the ith pelican in the jth dimension, I is a random number (1/2), p_j is the location of the prey in the jth dimension, and F_p is the objective function value. The effective updating allows the algorithm to prevent the move to non-optimal areas. This characteristic is described as:

$$X_i = \begin{cases} X_i^{P_1}, F_i^{P_1} < F_i \\ X_i, \text{else} \end{cases} \tag{7}$$

$X_i^{P_1}$ is the new status of the ith pelican and $F_i^{P_1}$ is its objective function. To assure the convergence to a better solution, the algorithm examines and scans the points in the neighbourhood of the pelican location. This process is simulated mathematically as:

$$x_{i,j}^{P_2} = x_{i,j} + R \cdot \left(1 - \frac{t}{T}\right) \cdot (2 \cdot rand - 1) \cdot x_{i,j} \tag{8}$$

$x_{i,j}^{P_2}$ is the new status of the *i*th pelican in the *j*th dimension as a second phase. Effective updating accepts or rejects the new pelican position, which is modelled as follows:

$$X_i = \begin{cases} X_i^{P_2}, F_i^{P_2} < F_i \\ X_i, \text{else} \end{cases} \tag{9}$$

$X_i^{P_2}$ is the new status of the *i*th pelican, and $F_i^{P_2}$ is the objective function value.

POA has significant superiority over other algorithms in solving optimization problems, it has a high exploration power ineffective checking of the search space, and finding of the optimal area.

4 Proposed approach

In our proposed enhancement approach, image generation is a pre-required phase. The application of Text-to-image generative model leads to obtain the aimed medical data for the examination and analysis of our enhancement algorithm.

After obtaining the images, our enhancement method is represented in two steps:

- Step1. The estimation of the clip-limit β based on using POA (using Eq. (10)).
- Step2. The contrast enhancement utilizing CLAHE (using Eq. (2)).

The objective function that defines the efficiency of our proposed model is the fit relationship between β and the following performance parameters: Peak Signal to Noise Ratio (PSNR), Structural Similarity Index (SSIM), Correlation Coefficient (CoC), Mean Square Error (MSE), Entropy (EL), and Standard Deviation (SD). After investigating the variation of β with these parameters, Eq. 2 can be presented as follows:

$$\begin{gathered}
\beta = 4,801.10^{-5}.\left[sin\left(X_1 - \pi\right)\right] + 1,732.10^{-5}\left[\left(X_1 - 10\right)^2\right] - 0,00361/X_1 \rightarrow PSNR \\
\beta = -41,88.\left[sin\left(X_2 - \pi\right)\right] + 1,427.\left[\left(X_2 - 10\right)^2\right] - 151/X_2 \rightarrow SSIM \\
\beta = -23,24.\left[sin\left(X_3 - \pi\right)\right] + 0.7188.\left[\left(X_3 - 10\right)^2\right] - 77,78/X_3 \rightarrow CoC \\
\beta = 8,993.10^{-5}.\left[sin\left(X_4 - \pi\right)\right] - 5,798.10^{-8}.\left[\left(X_4 - 10\right)^2\right] + 0,002474/X_4 \rightarrow MSE \\
\beta = 1,216.\left[sin\left(X_5 - \pi\right)\right] - 0,1245.\left[\left(X_5 - 10\right)^2\right] + 1,934/X_5 \rightarrow EL \\
\beta = -0,01488.\left[sin\left(X_6 - \pi\right)\right] + 2,2.10^{-5}.\left[\left(X_6 - 10\right)^2\right] - 0.03428/X_6 \rightarrow SD
\end{gathered} \tag{10}$$

Equation (10) has been fitted after performing several tests. The study of the distribution of the clip-limit and the performance parameters as dependent variables has been carried out. In this regard, the estimation of β appropriate value is essential to obtain a highly enhanced performance. By presenting the enhancement issue as an optimization problem, the novel nature-inspired algorithm POA is applied to improve the efficiency of the entire operation. POA is utilized in solving optimization issues in a variety of engineering disciplines, and it was first introduced for image-processing applications in this work.

CLAHE-POA Algorithm

/// STEP1 Text-to-image model.
1. **Input:** Original medical image generated by Text-to-image model.

/// STEP2 CLAHE
2. CLAHE enhancement function.

/// STEP3 POA
3. Input the objective function (optimization problem information using Eq. 10).
4. Determination of the population size (N) and the number of iterations.
5. Initialization of pelicans' position and the calculation of the objective function.
6. For $t=1:T$
7. Generation of the prey's position at random.
8. For $I=1:N$
9. Phase 1, 2: exploration and exploitation phases.
10. For $j=1:m$
11. New status' calculation of the jth dimension.
12. End.
13. End.
14. Updating the ith population member.
15. End.
16. Update and output the best solution obtained by POA.
17. Estimated value of β.

///
18. **Output:** Enhanced image.

5 Results

Experiments on medical images generated using the Text-to-image stable diffusion model have been made. The experimental data of different medical imaging modalities are saved in TIFF format (Ct-Scan Brain image, MRI Brain image, PET image, Ultrasound Kidney image). In addition, Axial CT-scan and MRI brain frames gathered from radiopedia [33, 34], which were utilized for a comparison process. The objective of using different types of experimental data is to qualitatively exhibit the enhancement performance of our model. The CLAHE-POA and Text-to-image generative algorithms are implemented using MATLAB (version 9.4) and PYTHON (google Colab) respectively.

In the experiments, in order to present the proposed enhancement method, and to illustrate the advantages of the amalgamation of CLAHE and POA, the performance of our proposed method was compared with eight state-of-the-art experimental methods. This includes the following: Wiener Filter WF [35], Gaussian Filter GF [36], Median Filter MF [37], Quantum Particle Swarm Optimization algorithm QPSO [38], Artificial Bee Colony Algorithm ABC [39], Unsharp Masking Algorithm UM [40], CSDNET model [41], and FilterNet model [42]. Besides, to verify the image enhancement comprehensively and objectively, the performance parameters are adopted. PSNR presents the enhancement performance objectively, larger values represent superior enhancement; SSIM defines the distortion of the enhanced images; CoC and EL indicate the image information and details;

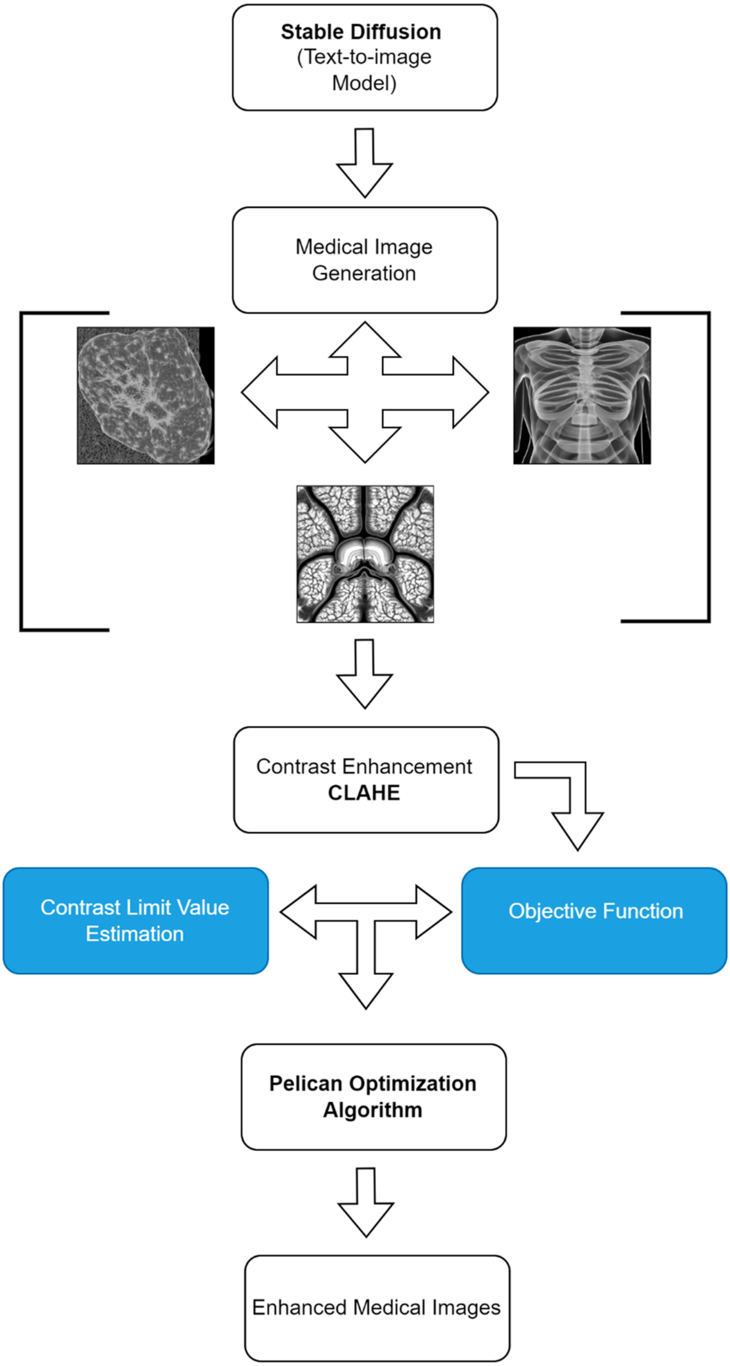
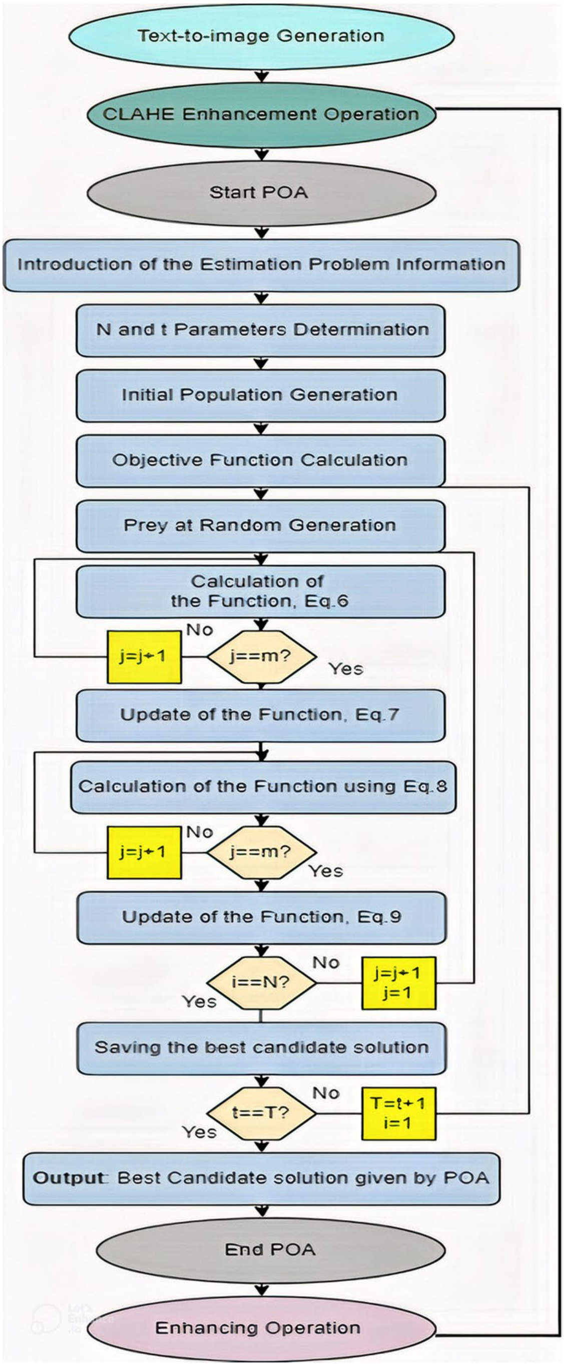


Fig. 3 The steps of the image enhancement operation

Fig. 4 The graphical block diagram of our proposed model



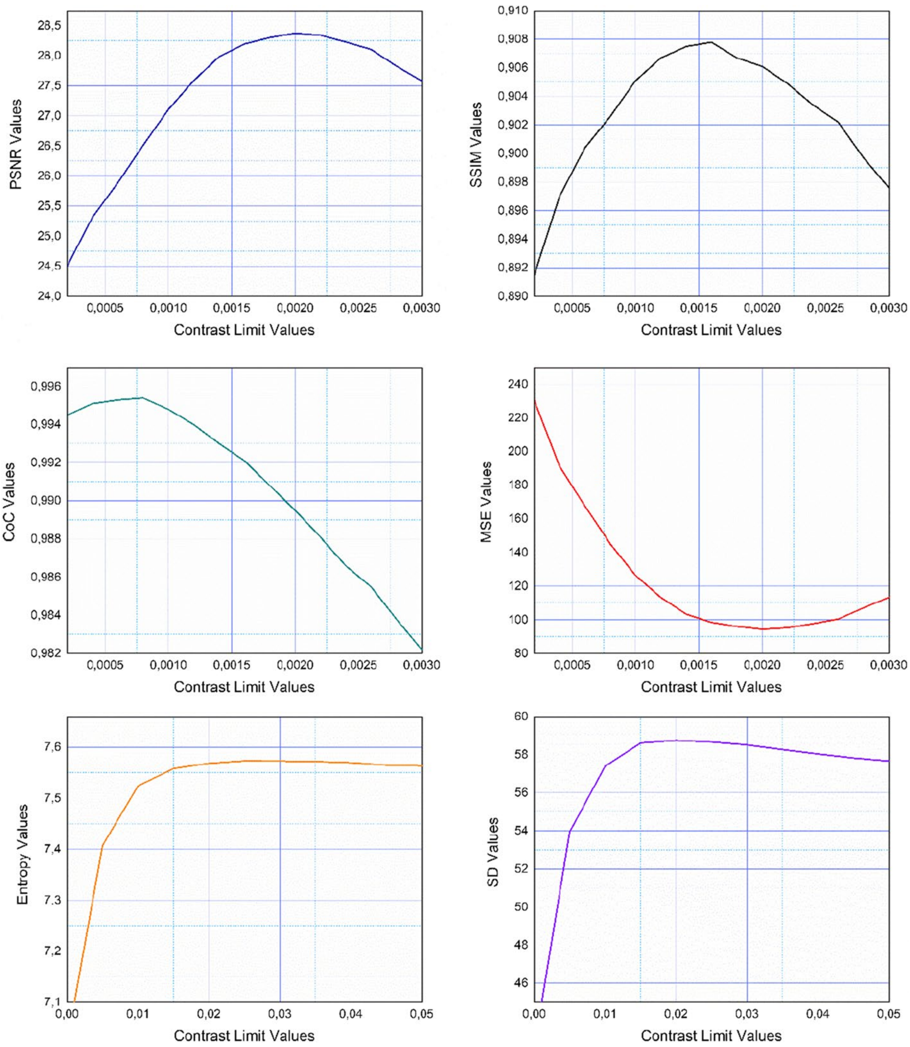


Fig. 5 The variation of the clip limit β with the performance parameters

and MSE and SD can present the contrast of the enhanced images, While MSE values decrease with the increasing of the overall contrast.

6 Discussion

Figure 3 represents the distribution of β in parallel with the performance parameters. On this basis, β increases in proportion with the parameters 'values, except MSE values which are inversely proportional with β and the overall contrast. The cut-off value of each curve indicates that β estimation is essential to achieve a superior enhancement performance (Figs. 4 and 5).

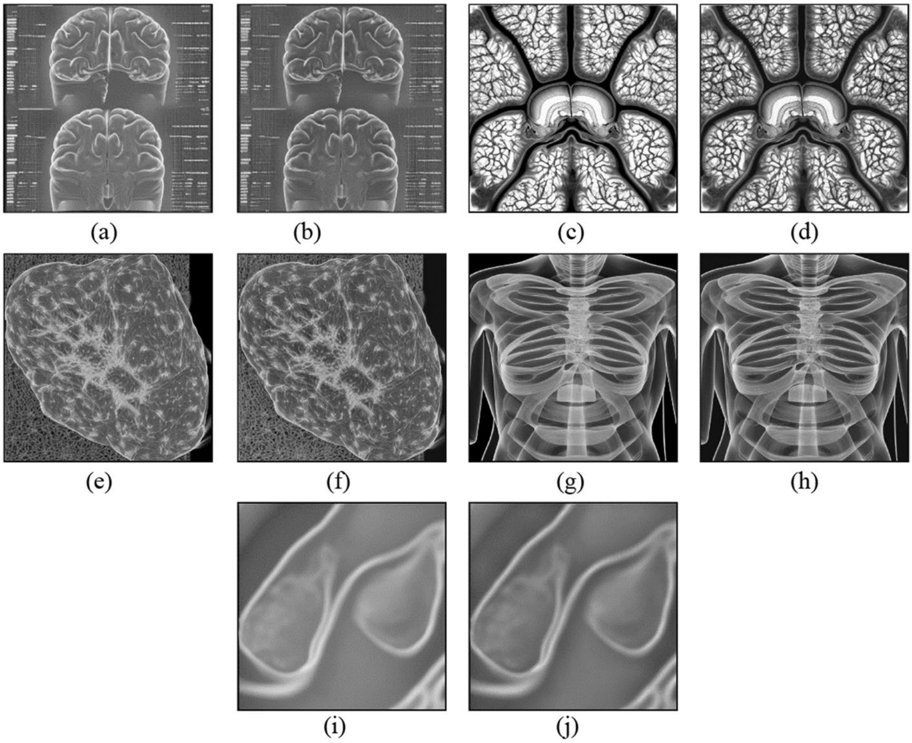


Fig. 6 Image enhancement of medical images (different imaging modalities) using our proposed model, **a** Original CT brain image. **b** Enhanced image, **c** Original MRI brain image, **d** Enhanced image, **e** Original PET image, **f** Enhanced image, **g** Original radiography chest image, **h** Enhanced image, **i** Original Ultrasound kidney image, **j** Enhanced image

The present study's most significant challenge involved determining the objective function that represents the fit relationship between β and the performance parameters. If the fit relationship is inappropriate, the enhancement performance would be less efficient. Applying a fitting tool with a high accuracy and precision characteristics would enhance even more the performance of our model.

Regarding visual comparisons, Figs. 6 and 7 show the enhancement results of generated medical images obtained by the present method and the other experimental state-of-the-art algorithms. By comparing details and local regions in the images, particularly regions with specific structures and shapes, we conclude that our proposed model offers superior visual quality. Edges and sharpness are blurred in CLAHE/GF and CLAHE/WF, but they are more preserved using our method. CLAHE/GQPO and CLAHE/ABC performances are close to some extent to our model in terms of visual quality. However, β estimation is more accurate using our algorithm, which resulted in a superiority of our method in terms of performance.

The results provided in Fig. 7, which represent the enhancement of a radiography chest image, proved the superiority of our model compared with the other algorithms. CLAHE/WF, CLAHE/GF, and CLAHE/MF over-increase the image brightness, while

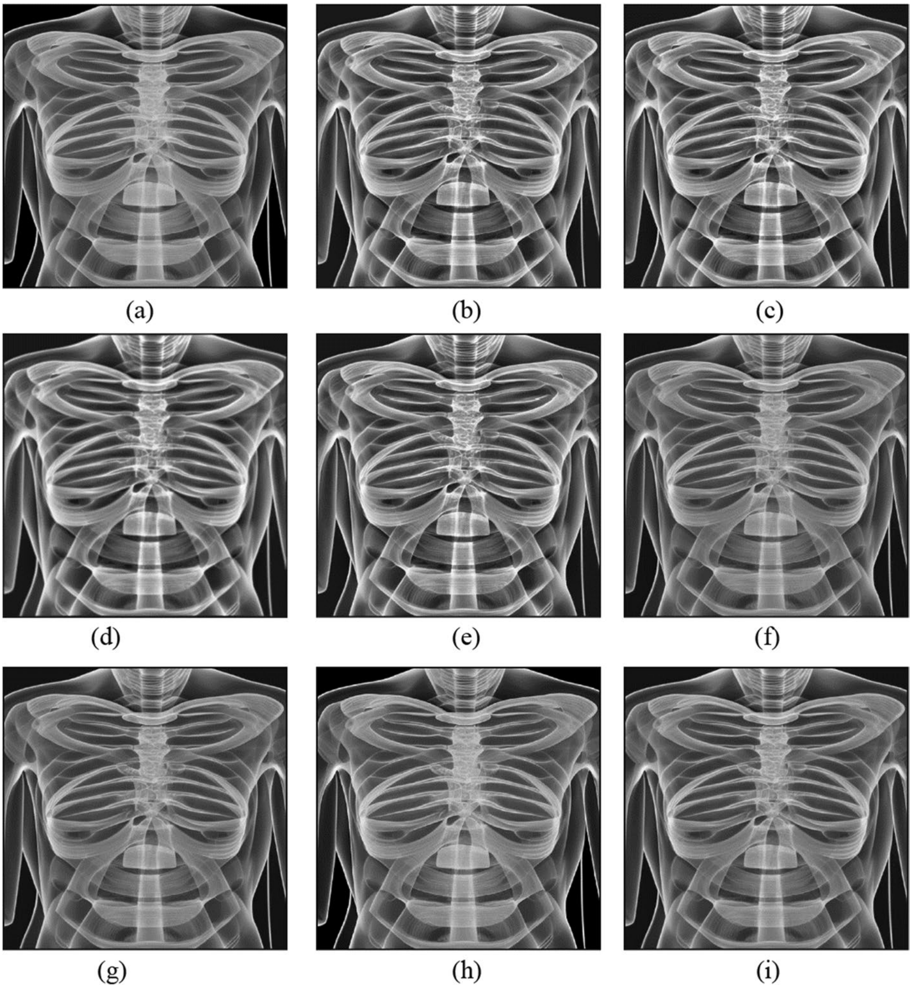


Fig. 7 Comparison of the enhancement process using our proposed model and other experimental methods. **a** Original radiography chest image, **b** Enhanced image using CLAHE, **c** Enhanced image using CLAHE/WF, **d** Enhanced image using CLAHE/GF, **e** Enhanced image using CLAHE/MF, **f** Enhanced image using CLAHE/QPSO, **g** Enhanced image using CLAHE/ABC, **h** Enhanced image using UM, **i** Enhanced image using our proposed model

the results of UM and our proposed algorithm are almost identical, with a slight advantage of our method in the overall contrast.

Concerning Quantitative evaluation, Fig. 8; Table 1 present the comparison of the numerical results based on the performance-relevant parameters. Our enhancement algorithm significantly outperforms the other experimental methods. In terms of PSNR values, our model and UM results are quite similar, with the superiority of our algorithm in comparison with all the other methods regarding the performance parameters SSIM, CoC, EL, and SD. Concerning MSE, UM results are preferable.

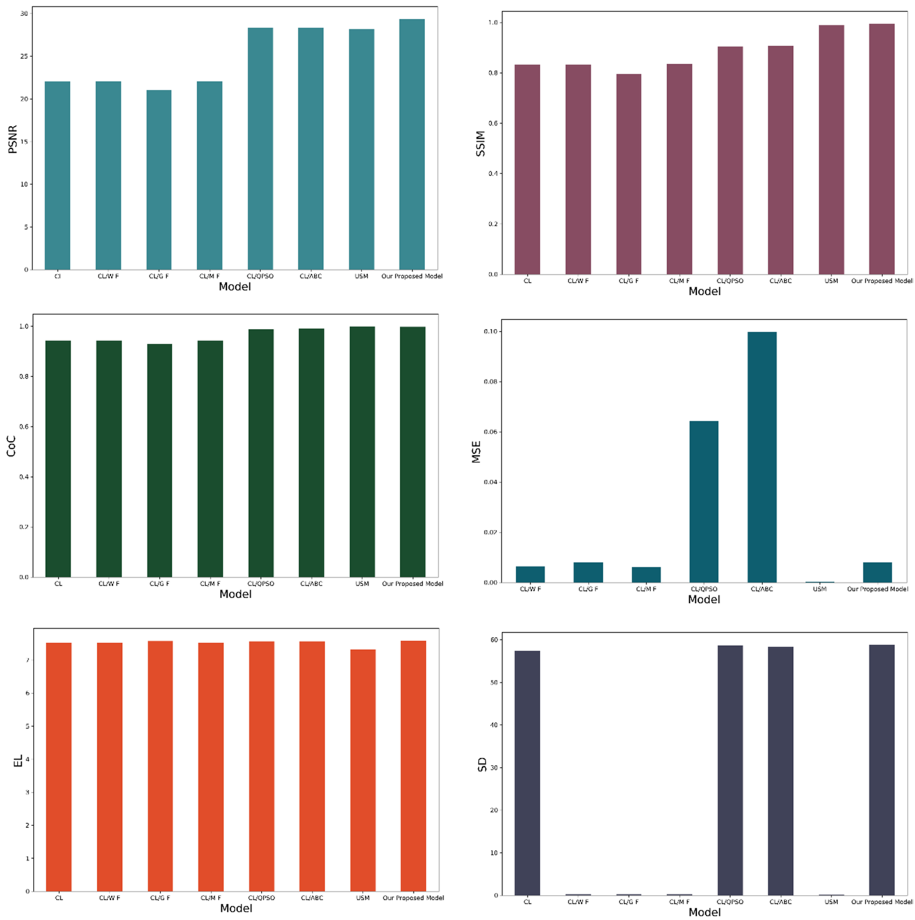


Fig. 8 Comparison of the enhancement efficiency using our proposed model and other experimental methods

In order to qualitatively describe the enhancement performance of our model, we applied our proposed algorithm in real medical image enhancement as shown in Fig. 9, which represents the enhancement of two axial brain images of different imaging modalities CT-Scan and MRI. Our method evidenced its efficiency in terms of enhancing performance, in comparison with recent enhancement methods, including CDSNET and FilterNet Enhancement models as shown in Fig. 10; Table 2. In addition to that, our method proves its applicability in various images. In this respect, improving the contrast and sharpness using our method gives a superior visual effect on the processed image and leads to a typical observation and diagnosis by specialists.

Table 1 Comparison of the enhancement efficiency of radiography chest image using our proposed model and the other experimental methods based on the performance parameters

Performance parameter	CLAHE	CLAHE/WF	CLAHE/GF	CLAHE/MF	CLAHE/QPSO	CLAHE/ABC	UM	Our proposed model
PSNR	22.023	22.034	22.04	22.071	28.37	28.353	28.171	29.347
SSIM	0.832	0.796	0.835	0.905	0.905	0.906	0.988	0.994
CoC	0.943	0.943	0.929	0.943	0.988	0.99	0.998	0.997
MSE	408.07	0.006	0.007	0.006	0.064	0.099	0.0002	0.008
EL	7.524	7.53	7.572	7.527	7.569	7.572	7.314	7.5822
SD	57.398	0.225	0.221	0.224	58.668	58.402	0.217	58.806

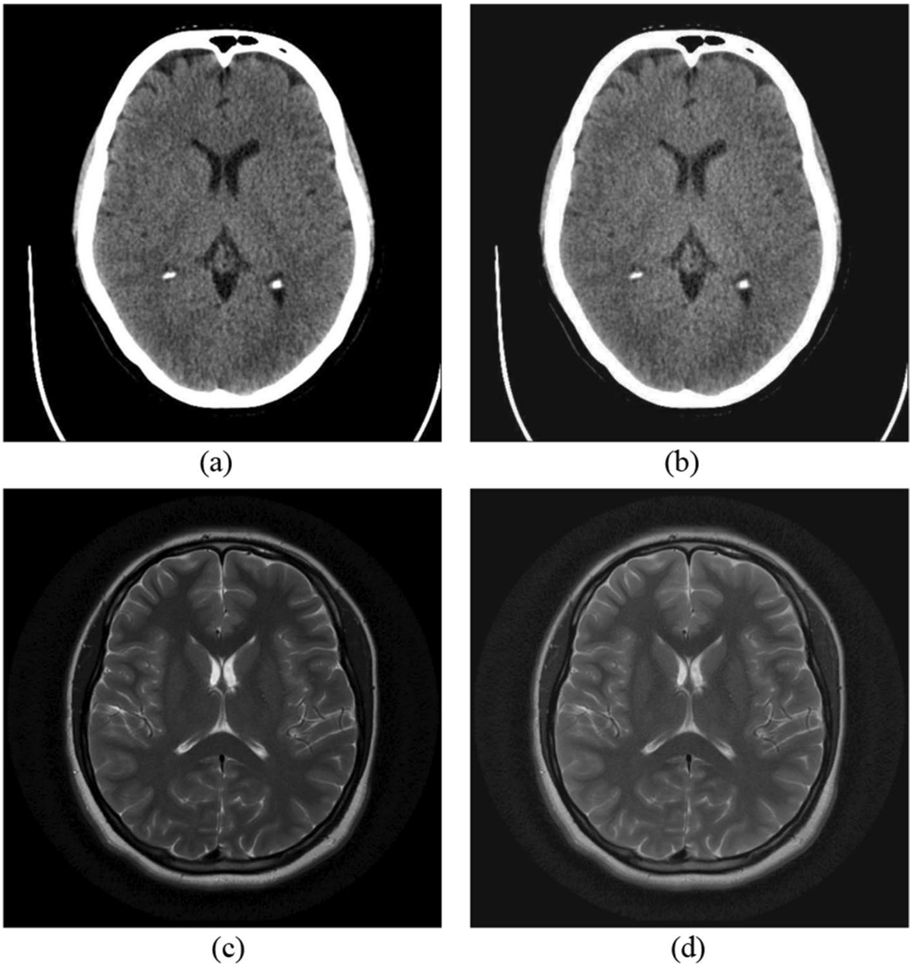


Fig. 9 CT and MRI real image enhancement using our proposed model. **a** Original axial CT-Scan brain image, **b** Enhanced image, **c** Original axial MRI brain image, **d** Enhanced image

7 Conclusion

Based on the amalgamation of both contrast adaptive histogram equalization and pelican optimization algorithm, our novel medical image enhancement method is presented. Through the estimation of the clip-limit as a solution to the optimization problem relating it to the performance parameters, the medical images' contrast is significantly improved. The experimental results prove the superiority of our model in comparison with the other state-of-the-art methods qualitatively and quantitatively, after conducting an in-depth examination and comparative analysis of performance parameters. Our proposed method is able to illustrate the structure and forms relevant details, which are contained in the medical images. All these steps lead to the increase of the overall contrast on the one hand and enhance the visual perception and observation on the other hand.

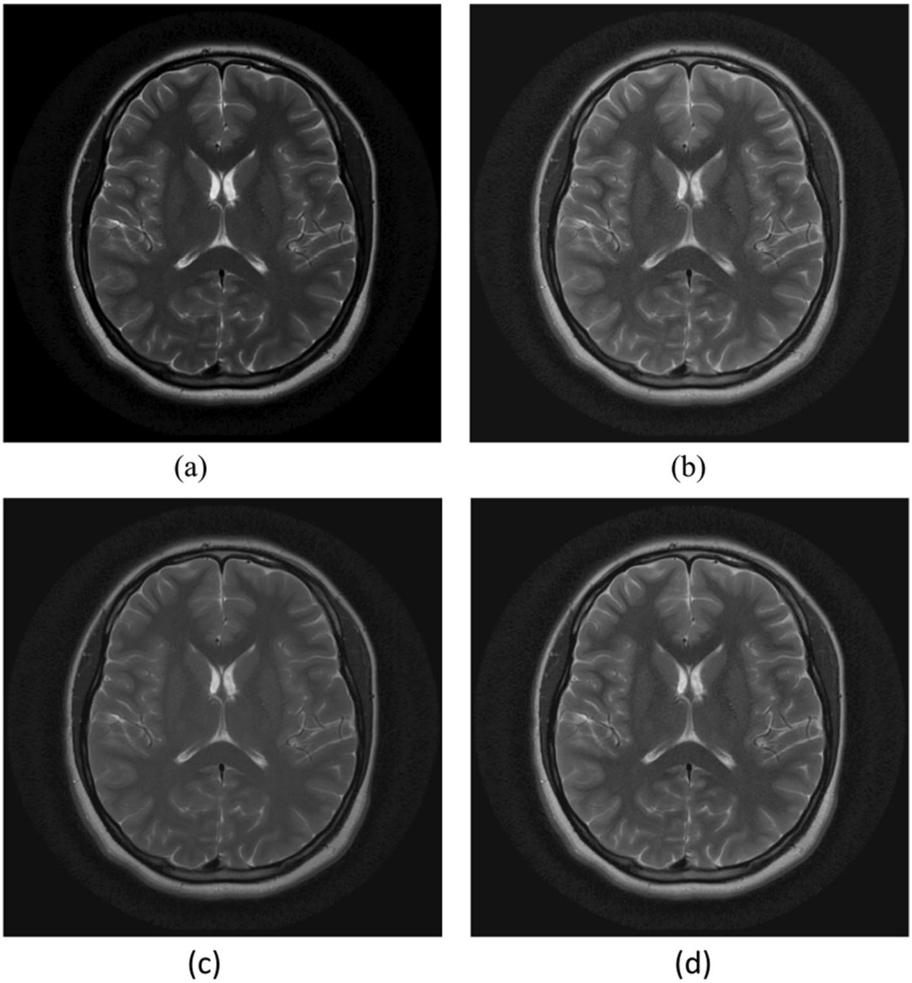


Fig. 10 Comparison of the enhancement performance on real medical images using our proposed model and other experimental methods. **a** Original axial MRI brain image, **b** Enhanced image using CSDNET, **c** Enhanced image using FilterNet, **d** Enhanced image using our proposed model

Table 2 Comparison of the enhancement efficiency of axial MRI brain image using our proposed model and the other experimental methods based on the performance parameters

Performance parameter	CSDNET	FilterNet	Our proposed model
PSNR	21.253	26.596	29.256
SSIM	0.812	0.985	0.989
CoC	0.97	0.98	0.992
MSE	0.00015	0.0001	0.0075
EL	7.135	7.425	7.5647
SD	0.568	0.217	57.265

On the basis of applying POA to estimate the clip-limit, which controls the performance of the enhancement operation by utilizing CLAHE. The estimation process improves the efficiency of the operation and provides superior results regarding the image quality and contrast. Improving the quality of the medical images using our algorithm allows the achievement of a superior visual impact on the processed image and increases the rate of conformity in the clinical diagnosis. For future works, the principle of this model can be expanded to other image-processing issues, particularly medical image segmentation, and denoising applications.

Funding The authors had received no funding.

Data availability Dataset used would be provided by the corresponding author upon a reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare that they have no competing interests.

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